Blood Bank Inventory & Emergency Donor Matcher

Jaanhavi Ashwin Kher | PES2UG24CS199 | 20/08/2026

Deliverable - 1

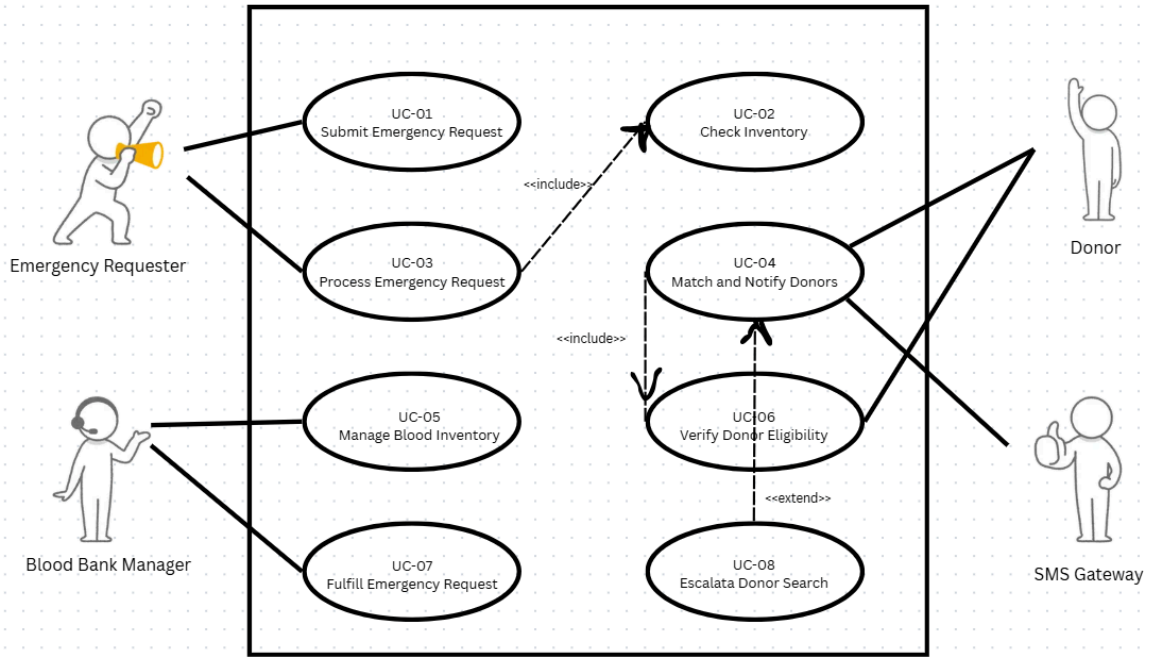
Blood Bank Inventory & Emergency Donor Matcher — Requirements Table

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Pass: Compatible donors notified within 30 seconds of emergency flag. Fail: Incompatible blood group alerted.	Core emergency-response function; speed and accuracy of donor matching directly affects patient survival.
FR-002	Functional	The system shall allow the Blood Bank Manager to add, update, and remove blood unit records in inventory, capturing blood group, component type, collection date, and expiry date.	High	Pass: A new/updated blood unit appears in the inventory ledger within 5 seconds with all mandatory fields populated. Fail: Record saved with a missing blood group, component type, or expiry date.	Accurate, up-to-date inventory is the foundation for every matching and allocation decision.
FR-003	Functional	The system shall automatically flag blood units nearing expiry (within 24 hours) and exclude expired units from the allocatable stock shown to matching and search functions.	Medium	Pass: Units within 24 hours of expiry are labelled "Near-Expiry" and units past expiry no longer appear in available-stock queries. Fail: An expired unit is returned as available or allocated to a request.	Prevents unsafe transfusions and reduces wastage by prompting timely use or disposal of ageing stock.
FR-004	Functional	The system shall allow donors to register and maintain a profile (blood group, contact number, geolocation, last donation date) and shall verify donor eligibility (minimum 90 days since last donation) before including them in any emergency match.	High	Pass: A donor whose last donation was less than 90 days ago is excluded from match results. Fail: An ineligible donor receives an emergency SMS alert.	Protects donor health and ensures only medically eligible donors are contacted during emergencies.
FR-005	Functional	The system shall allow an Emergency Requester to submit a blood request specifying blood group, component, quantity, and urgency level, and shall generate a unique, trackable Request ID on submission.	High	Pass: Submitting a valid request returns a unique Request ID and sets status to "Open" within 3 seconds. Fail: A request is accepted without a blood group, quantity, or urgency level.	A structured, uniquely identified request is required before the system can search inventory or trigger donor matching.
NFR-001	Nonfunctional	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Concurrent allocation of a single unit to two requests could cause a fatal shortfall at the point of transfusion.
NFR-002	Nonfunctional	The system shall encrypt all donor and patient personal/health data at rest and in transit, and restrict access using role-based permissions (Manager, Requester, Donor).	High	Pass: Penetration test confirms no plaintext PII/health data is exposed at rest or in transit, and users can access only functions permitted by their role. Fail: Any donor/patient record is retrievable by a role without explicit access rights.	Blood group and health data are sensitive personal information; a breach would violate donor trust and data-protection regulations.

Deliverable - 2

Blood Bank Inventory and Emergency Donor Matcher



Deliverable - 3

Use Case: UC-04 — Match & Notify Emergency Donors

Primary Actor	System (triggered on behalf of Emergency Requester)
Secondary Actors	Emergency Requester, Blood Bank Manager, Donor, SMS Gateway
Preconditions	1) A valid emergency blood request has been submitted via UC-01 — Submit Emergency Request and processed via UC-03 — Process Emergency Request (which includes UC-02, Check Inventory). 2) The blood bank inventory ledger is online and reachable. 3) At least one donor profile exists in the system with geolocation data.
Postconditions	Success: Compatible, eligible donors within range have been notified by SMS and the request status is updated to "Donors Notified." Failure: The request status is updated to "No Match Found" and the Blood Bank Manager is alerted for manual escalation.

Main Success Scenario

1. The system receives a validated emergency Request ID from UC-03 (Process Emergency Request), containing blood group, component, quantity, and urgency level, along with the confirmed inventory shortfall from UC-02 (Check Inventory).
2. The system verifies donor eligibility for the requested blood group («include» "Verify Donor Eligibility", UC-06; FR-004), checking that each candidate donor's last donation was at least 90 days ago.
3. The system compiles the list of eligible donors returned by UC-06 and filters them to those within a 10 km radius of the requesting facility, using stored donor geolocation.
4. The system broadcasts an SMS alert to every matching donor via the SMS Gateway, including blood group needed, facility location, and urgency (FR-001).
5. The system logs the notification timestamp against the Request ID and opens a 15-minute response window during which alerted donors can confirm availability.
6. A responding Donor confirms availability directly with the system before the window closes; the system marks the donor as "Responding" and notifies the Blood Bank Manager.
7. The Blood Bank Manager reviews responding donors and confirms an appointment with one or more of them through UC-07 (Fulfill Emergency Request), updating the request status to "Donors Notified / In Progress."
8. Use case ends successfully.

Alternate Flow

2a. Too Few Eligible Donors Found

- **2a1.** UC-06 (Verify Donor Eligibility) returns fewer eligible donors than the request quantity requires within the initial candidate pool.
- **2a2.** The system triggers UC-08 (Escalate Donor Search), which «extend»s UC-06 by widening the eligibility search — first to donors just outside the standard radius, then in 5 km increments up to a configured maximum.
- **2a3.** If additional eligible donors are found at the wider search, the flow resumes at Main Success Scenario step 3 with the expanded donor set.
- **2a4.** If still no eligible donors are found at maximum radius, the system sets the request status to "No Match Found" and immediately alerts the Blood Bank Manager to initiate manual escalation (e.g., contacting partner blood banks).

5a. Response Window Expires With No Confirmation

- **5a1.** No alerted donor confirms availability before the 15-minute response window closes.
- **5a2.** The system automatically re-triggers UC-08 (Escalate Donor Search) to notify additional eligible donors at a wider radius.
- **5a3.** The system logs the timeout against the Request ID and notifies the Blood Bank Manager that the initial round produced no response, so the flow rejoins Main Success Scenario step 4 with the expanded donor set.